

Acute Responses to Repeated-Sprint Training in Hypoxia Combined With Whole-Body Cryotherapy: A Preliminary Study

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Purpose: This study aimed to investigate acute psychophysiological responses to repeated-sprint training in hypoxia (RSH) combined with whole-body cryotherapy (WBC). **Method:** Sixteen trained cyclists performed 3 sessions in randomized order: RSH, WBC-RSH (WBC pre-RSH), and RSH-WBC (WBC post-RSH). RSH consisted of 3 sets of 5 × 10-second sprints with 20-second recovery at a simulated altitude of 3000 m. Power output, muscle oxygenation (tissue saturation index), heart-rate variability, and recovery perception were analyzed. Sleep quality was assessed on the nights following test sessions and compared with a control night using nocturnal ActiGraphy and heart-rate variability. **Results:** Power output did not differ between the conditions ($P = .27$), while the decrease in tissue saturation index was reduced for WBC-RSH compared to RSH-WBC in the last set. In both conditions with WBC, the recovery perception was higher compared to RSH (WBC-RSH: +15.4%, and RSH-WBC: +21.9%, $P < .05$). The number of movements during the RSH-WBC night was significantly lower than for the control night (−18.7%, $P < .01$) and WBC-RSH (−14.9%, $P < .05$). RSH led to a higher root mean square of the successive differences of R-R intervals and high-frequency band during the first hour of sleep compared to the control night ($P < .05$) and RSH-WBC ($P < .01$). **Conclusions:** Inclusion of WBC in an RSH session did not modify the power output but could improve prolonged performance in hypoxia by maintaining muscle oxygenation. A single RSH session did not deteriorate sleep quality. WBC, particularly when performed after RSH, positively influenced recovery perception and sleep.

Keywords: athletes, physical performance, cycling, recovery, sleep

The use of environmental stressors (eg, altitude,¹ heat, and cold²) has become increasingly common to improve the physical performance and recovery of elite athletes. Altitude/hypoxia training has long been used exclusively in endurance sports. Nevertheless, for the past 10 years, innovative hypoxia training methods, such as repeated sprint training in hypoxia (RSH), have been used for endurance sports,³ but also activities for which the ability to repeat efforts is an important factor in performance, as in the case of team sports⁴ or racket sports.⁵

RSH is defined as the repetition of maximal efforts (≤ 30 s) with incomplete recoveries (≤ 60 s) performed in hypoxia.^{6,7} Compared to repeated sprint training in normoxia (RSN), RSH induces repeated sprint ability improvement with enhanced muscle blood perfusion, and delaying the onset of fatigue and exhaustion.⁴ In contrast with RSN, the performance improvements after RSH can be attributed to increased environmental stress experienced by athletes. Maldonado-Rodriguez et al⁸ have highlighted an increase in heart rate (HR), blood lactate concentration ([La[−]]), and rating of perceived exertion (RPE) related to a decrease in oxygen saturation, along with a lower oxygen availability at the muscle level during RSH.⁹ Past studies have also recorded a performance impairment throughout RSH (vs RSN),^{8,9} as well as an increase in peripheral fatigue indices. The field of research into

RSH is now broad, with studies of different exposure altitudes, work-to-rest ratios, but also combined with other environmental stresses.^{10,11}

Optimal recovery is paramount to prevent fatigue accumulation and the risk of injury.¹² Among the recovery methods commonly used, whole-body cryotherapy (WBC) is characterized by short whole-body (from head to toe) exposure to dry and intense cold.² WBC causes a sudden reduction in skin temperature, inducing chain reactions initiated by the stimulation of thermoreceptors. These reactions send a message to the hypothalamus and trigger adaptation responses to cold, such as peripheral vasoconstriction and activation of the sympathetic system.^{2,13} The peripheral blood redirection to the core of the body stimulates baroreceptors and leads to a subsequent diminution of the sympathetic nerve activity and an increase in vagal control after WBC.² An effect on the autonomic nervous system was also observed, improving resting postexercise heart rate variability (HRV),^{13,14} and peripheral vasodilatation that may persist for several hours.² Cold-induced peripheral vasodilatation contributes to faster recovery of physiological and psychological parameters, such as muscle oxygenation and RPE during subsequent exercises, compared to passive recovery.¹⁵ WBC also enhances sleep quality, with decreased movement, parasympathetic nervous activity improvement during the night and a better perception of sleep quality.^{16,17}

WBC exposure prior to exercise has demonstrated several positive effects.¹⁸ Among these effects, Krüger et al¹⁵ linked increased muscle oxygenation after WBC exposure with time-trial performance improvement and cardiovascular strain reduction

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during following exercise. Muscle oxygen availability played an important role in maintaining performance in hypoxia.⁹ Consequently, in a competitive context, preexercise exposure to WBC could be useful in maintaining a high level of performance, close to that achieved at sea level, even if the athlete has not been able to acclimatize to the altitude.

Several beneficial adaptations have also been reported for WBC performed after exercise.² To our knowledge, there is a lack of data on the effects of combining RSH and WBC. Therefore, this study aimed to determine whether including WBC prior (WBC-RSH) or after (RSH-WBC) RSH session would modify the physiological responses (power output, HR, $[La^-]$, and muscular oxygenation) during exercise and subsequent recovery (HRV indices, sleep quality, and perceived recovery). In the present study, we tested the following hypotheses: WBC-RSH would improve performance during exercise in hypoxia compared with RSH and RSH-WBC, through an increase in muscle oxygenation¹⁵; RSH-WBC would induce improved sleep compared to RSH and control night^{16,17}; and both WBC conditions would enhance recovery of HRV indices compared to RSH.^{13,14}

Methods

Participants

Sixteen males (age: 21 [3] y, height: 178.8 [6.5] cm, body mass: 70.4 [7.8] kg and body fat: 11.8% [2.8%]), including 6 high-level cyclists (11 [1] h·wk⁻¹ of training and competing at national level) and 10 trained cyclists (7 [3] h·wk⁻¹ of training and competing at regional level),¹⁹ volunteered to take part in the experiment.

Experimental Design

This study used a crossover experimental design to determine the combined effects of RSH and WBC on performance and recovery. The experiment comprised 5 sessions, separated by 7 days. The participants' first visit involved a medical examination, including an electrocardiogram and clinical inspection for contraindications to hypoxia and WBC. After their inclusion, each participant familiarized themselves with WBC.

The second visit involved familiarization with repeated sprint effort completed in normoxia. It consisted of a 10-minute warm-up and 2 sets of five 10-second all-out sprints, separated by 20 seconds of active recovery. The resistance for each training period was the same as that used during the test sessions. The third, fourth, and fifth visits were test sessions to evaluate the effect of RSH and WBC combination. Three conditions were tested in a randomized order for every participant: RSH, WBC-RSH, and RSH-WBC.

Figure 1 presents the experimental design of the sessions. The test sessions started with an HRV measurement performed for 5 minutes in the supine position (HRV₁). Depending on the experimental condition, participants were subsequently exposed to WBC (condition: WBC-RSH) or thermoneutral stress (conditions: RSH and RSH-WBC). WBC was conducted in a Cryowind chamber (Aurore Concept). Participants were exposed to ventilated cold for 30 seconds at -25 °C in the first chamber and then for 3 minutes at -60 °C in the second chamber.²⁰ Participants were exposed to cold with minimal clothing (protection on their feet, hands, and ears). Exposure to thermoneutral stress was performed in a sitting position for 10 minutes at 20 °C.

A second HRV measurement (HRV₂) was performed using the same modalities as HRV₁. Then, RSH was performed on a mechanical resistance bicycle. It consisted of a 10-minute warm-up (resistance: 0.2 N·m·kg⁻¹), followed by 3 sets of five 10-second all-out sprints (resistance: 0.6 N·m·kg⁻¹) separated by 20 seconds of active recovery (resistance: 0.0 N·m·kg⁻¹). An active recovery of 5 minutes was implemented between the sets (resistance: 0.2 N·m·kg⁻¹). Each training session ended with an active recovery of 10 minutes. The following variables were recorded: power output, HR, muscle oxygenation, oxygen saturation, lactatemia, and RPE measurements. RSH was performed in a tent using a hypoxic generator (Cloud 9 Altitude, Sporting Edge LTD) to simulate hypoxic stress equal to 3000 m³ (fraction of inspired oxygen = 0.147). Upon exiting the tent after RSH, a new HRV measurement was taken (HRV₃), followed by WBC (RSH-WBC condition) or recovery in a sitting position in a thermoneutral room (for the other 2 conditions). Finally, the last HRV measurement was taken at the end of the protocol (HRV₄).

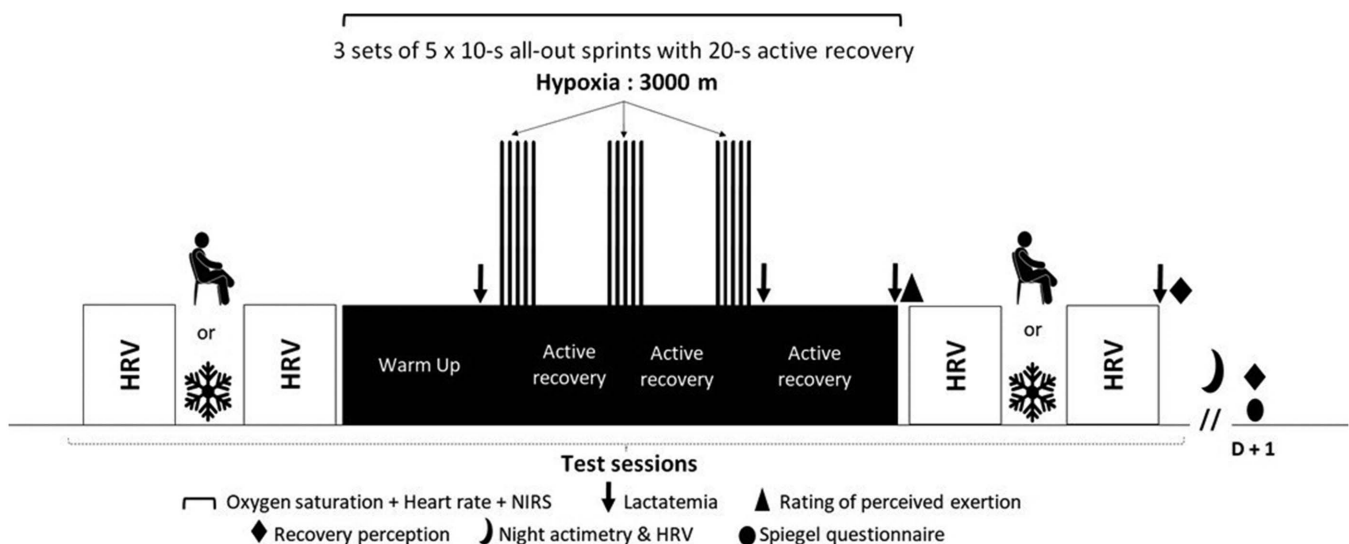


Figure 1 — Schematic representation of the test sessions. D + 1 indicates day after the test session; HRV, heart-rate variability; NIRS, near-infrared spectroscopy.

Measurements

Power Output

Power output was recorded with an SRM power sensor (Schoberer Rad Messtechnik) at a sampling rate of 1 Hz. The mean power output throughout the session and for each sprint was analyzed. The decrement score (S_{dec}) was calculated for every training set as follows²¹:

$$S_{\text{dec}, \%} = 1 - (\text{sum of mean power output over 5 sprints}) / (\text{best mean power output} \times 5) \times 100.$$

Oxygen Saturation

The participants' blood oxygen saturation (SpO_2) was monitored during hypoxia exposure using a pulse oximeter (Berry Med BM2000A, Berry Electronic Technology Co) placed on their left index finger at a sampling rate of 1 Hz. SpO_2 was averaged over the entire session.

HR and HRV

HR was recorded throughout the training using Polar H10 (Polar Electro Oy). The mean HR over the entire session was computed and then decomposed for each training set. During the HRV measurement, the participants were in a supine position in a quiet environment. HRV measurement was performed for 5 minutes using Elite HRV application connected via Bluetooth to the HR monitor (Elite HRV LLC).²² The HRV data were registered at a sampling rate of 1000 Hz, and the R-R intervals were analyzed using Kubios HRV analysis software (version 3.5.0, Biosignal Analysis and Medical Imaging Condition, University of Kuopio). The last 4 minutes of the recording were considered for the analysis. Artifact noise was automatically replaced with the interpolated adjacent R-R interval value by applying a medium threshold of correction. The following time-domain-dependent HRV indices were analyzed: the R-R intervals (RR, ms), the root mean square of the successive differences of the normal R-R intervals (RMSSD, ms), and the SD of the normal-to-normal interval (ms). Instantaneous beat-to-beat variability (SD1, ms) and continuous beat-to-beat variability (SD2, ms) derived from the Poincaré plot analysis were analyzed. The frequency domain was investigated using low- and high-frequency spectral bands (respectively, ms^2). Among the HRV indices, RMSSD, SD1, and HF reflect cardiac parasympathetic activity, whereas RR, HR, SD of the normal-to-normal interval, SD2, and low-frequency spectral bands reflect both cardiac vagal and sympathetic activities.²³ For each index, the value recorded during HRV_1 was defined as the baseline at 100%. Variations from other measurements were calculated from this 100%.

Muscle Oxygenation

Muscle oxygenation was assessed using near-infrared spectroscopy with Oxymon Mk III (Artinis). To measure changes in muscle oxygenation, a double optode sensor was attached to the vastus lateralis of the participants' right quadriceps, following Faiss et al.³ Signals were recorded at a sampling frequency of 10 Hz. A 10th-order low-pass zero-phase Butterworth filter (cutoff frequency: 0.1 Hz) was utilized to reduce artifacts and smooth perturbations in the signal provoked by the pedaling. Oxyhemoglobin ($[\text{O}_2\text{Hb}]$) and deoxyhemoglobin ($[\text{HHb}]$) concentrations were monitored. Total hemoglobin ($[\text{tHb}]$) was calculated as the sum of $[\text{O}_2\text{Hb}]$ and

$[\text{HHb}]$, and was used as an index of the regional blood volume.²⁴ The tissue saturation index (TSI; expressed in percentages and calculated as $[\text{O}_2\text{Hb}]/([\text{tHb}])$) was used as an indicator of muscle oxygenation.^{10,15} Before the start of the RSH, a 1-minute stabilization period was carried out, during which participants sat on the saddle of the bicycle and kept their right-leg straight. The last 30 seconds of this stabilization period were averaged and taken as the resting values. Changes in $[\text{O}_2\text{Hb}]$, $[\text{HHb}]$, $[\text{tHb}]$, and TSI (Δ) were calculated and normalized from this stabilization period (arbitrarily defined as 0).²⁴ Maximal values of $[\text{HHb}]$, and minimum values of $[\text{O}_2\text{Hb}]$, $[\text{tHb}]$, and TSI, obtained at the end of each sprint were used to calculate Δ .

Lactatemia Measurements

Four capillary blood samples were drawn: at the end of the warm-up (ninth minute, $[\text{La}^-]_1$), 1 minute after the last sprint ($[\text{La}^-]_2$), just after the active recovery ($[\text{La}^-]_3$), and immediately after the last HRV_4 ($[\text{La}^-]_4$). Ear lobe blood samples (3 μL) were collected, and $[\text{La}^-]$ was analyzed using an automatic analyser, Lactate Pro 2 (Arkray Inc).

Perceptual Responses

RPE data were collected at the exit of the hypoxic chamber using the CR10 scale.²⁵ Perceived recovery scores were provided by the participants using the Edwards perceptive scale²⁶ immediately after HRV_4 (D) and again upon waking up the next morning (D + 1).

Sleep Quality

Sleep quality was evaluated objectively and subjectively after each test session and 3 nights after the second visit. Only the last night after the second visit was considered the baseline (CONT); the first 2 nights were considered habituation. Activity during sleep was determined with an accelerometer measuring movements on horizontal, vertical, and perpendicular axes. Participants were asked to wear a wrist ActiGraph (WGT3XBT monitor) at least 2 hours before bedtime, during sleep and until 1 hour after waking up the next morning. Raw data were recorded at an epoch length of 60 seconds and a sampling rate of 100 Hz and extracted using ActiLife software (version 6.11.0). The parameters obtained were time in bed calculated as the total time spent in bed (minutes), total sleep time defined as the time spent asleep (minutes), the wake after sleep onset calculated as the time during which the number of movements was over the threshold of 40 counts per minute (minutes), the vector magnitude calculated as the square root of the sum of the square of acceleration for each of the 3 axes (counts per minute) and sleep efficiency calculated as the ratio of total sleep time and total time spent in bed multiplied by 100 (%).¹⁷

HRV measurement was also performed during the night with Polar H10 and Elite HRV application. The analysis started 30 minutes after the bedtime. The recordings were analyzed during the first hour of sleep with a medium threshold correction using Kubios HRV software. The nights were considered for analysis only when a subject presented 4 recordings with a duration higher than 1.5 hours. Thus, only 12 subjects were included in the night HRV analysis. CONT night was arbitrarily normalized to 100%. The HRV variations for other nights were calculated with respect to the baseline (CONT).

Subjective sleep quality was also assessed using the Spiegel sleep quality perception questionnaire.¹⁶

Statistical Analysis

The statistical analysis was performed using Jeffreys's Amazing Statistics Program (JASP software, version 0.16.4.0). The results were expressed as the mean and SD, unless otherwise specified. The differences between the groups were analyzed using a 2-way (condition $\times$ time) or a 1-way (group) analysis of variance (ANOVA) with repeated measures. Normality was checked for each variable using the Shapiro–Wilk test. For nonnormal variables, an equivalent nonparametric test was used. The assumption of data sphericity was assessed and corrected with the appropriate Greenhouse–Geisser or Huynh–Feldt correction. The level of statistical significance was set at $P < .05$. The Holm post hoc test was applied when the ANOVA revealed a significant interaction or main effects. The effect size was calculated as partial eta squared (η_p^2) for ANOVA with following threshold: $<.06$ small, $.06$ to $.14$ moderate and $>.14$ large; and Cohen d ([95% confidence interval]) for the Holm test. The following threshold values for Cohen d were adopted: <0.20 trivial, 0.20 to 0.49 small, 0.50 to 0.79 moderate, and >0.80 large.

Results

Power Output

The mean power output was similar for the 3 conditions ($2.1 [0.4]$, $2.1 [0.4]$, and $2.1 [0.4]$ $\text{W}\cdot\text{kg}^{-1}$ for RSH, WBC-RSH, and RSH-WBC, respectively; $P > .05$; Figure 2). S_{dec} that quantified fatigue during each set was not different for the 3 conditions.

HR and HRV

The mean HR during training was not impacted by the 3 conditions ($144.5 [13.8]$ $\text{beats}\cdot\text{min}^{-1}$, $144.0 [16.1]$ $\text{beats}\cdot\text{min}^{-1}$, and $146.2 [15.5]$ $\text{beats}\cdot\text{min}^{-1}$ for RSH, WBC-RSH, and RSH-WBC, respectively; $P > .05$, $\eta_p^2 = .063$). The mean HR for each training set is presented in Figure 3. During first set, a small effect was observed for WBC-RSH

compared to RSH-WBC, where mean HR tended to be lower without reaching significant level ($P = .09$, $d = -0.356 [-0.859 \text{ to } 0.146]$).

Variations in HRV indices relative to HRV_1 are summarized in Table 1. No effect was observed on HR or HRV due to different conditions. However, the condition $\times$ time interaction was significant for RR ($P = .002$), with a significant increase in RR in HRV_2 for WBC-RSH compared to RSH ($P < .001$, $d = 1.162 [0.061 \text{ to } 2.263]$) and RSH-WBC ($P = .004$, $d = 0.884 [-0.106 \text{ to } 1.874]$).

Muscle Oxygenation, Oxygen Saturation, and Lactatemia Measurement

Graphical representations of $\Delta [\text{O}_2\text{Hb}]$, $\Delta [\text{HHb}]$, $\Delta [\text{tHb}]$, and ΔTSI from stabilization period are presented in Figure 4. The ANOVA revealed that there was no difference between conditions in all Δ . Nevertheless, the condition $\times$ time interaction was significant for $\Delta [\text{O}_2\text{Hb}]$, $\Delta [\text{tHb}]$, and ΔTSI . WBC-RSH was associated with a lower $\Delta [\text{O}_2\text{Hb}]$ compared to RSH-WBC during set 3 at sprint 1 ($P = .065$, $d = 0.520 [-0.725 \text{ to } 1.764]$), sprint 2 ($P = .097$, $d = 0.482 [-0.745 \text{ to } 1.710]$), sprint 3 ($P = .067$, $d = 0.521 [-0.724 \text{ to } 1.765]$), sprint 4 ($P = .047$, $d = 0.553 [-0.707 \text{ to } 1.813]$), and sprint 5 ($P = .078$, $d = 0.499 [-0.736 \text{ to } 1.734]$). In same set, ΔTSI was reduced for WBC-RSH compared to RSH-WBC (sprint 1: $P = .053$, $d = 0.436 [-0.636 \text{ to } 1.507]$; sprint 2: $P = .066$, $d = 0.426 [-0.640 \text{ to } 1.493]$; sprint 3: $P = .054$, $d = 0.440 [-0.634 \text{ to } 1.513]$; sprint 4: $P = .054$, $d = 0.438 [-0.635 \text{ to } 1.510]$; sprint 5: $P = .084$, $d = 0.405 [-0.651 \text{ to } 1.461]$). The mean SpO_2 was not significantly different between the conditions ($82.9\% [4.2\%]$, $81.7\% [5.3\%]$, and $83.1\% [2.7\%]$ for RSH, WBC-RSH, and RSH-WBC, respectively; $P > .05$, $\eta_p^2 = .007$). Table 1 lists $[\text{La}^-]_1$, $[\text{La}^-]_2$, $[\text{La}^-]_3$, and $[\text{La}^-]_4$.

Perceptual Sensations

RPE was not different between RSH ($7.2 [0.4]$ arbitrary units), WBC-RSH ($7.1 [0.4]$ arbitrary units) and RSH-WBC (7.0

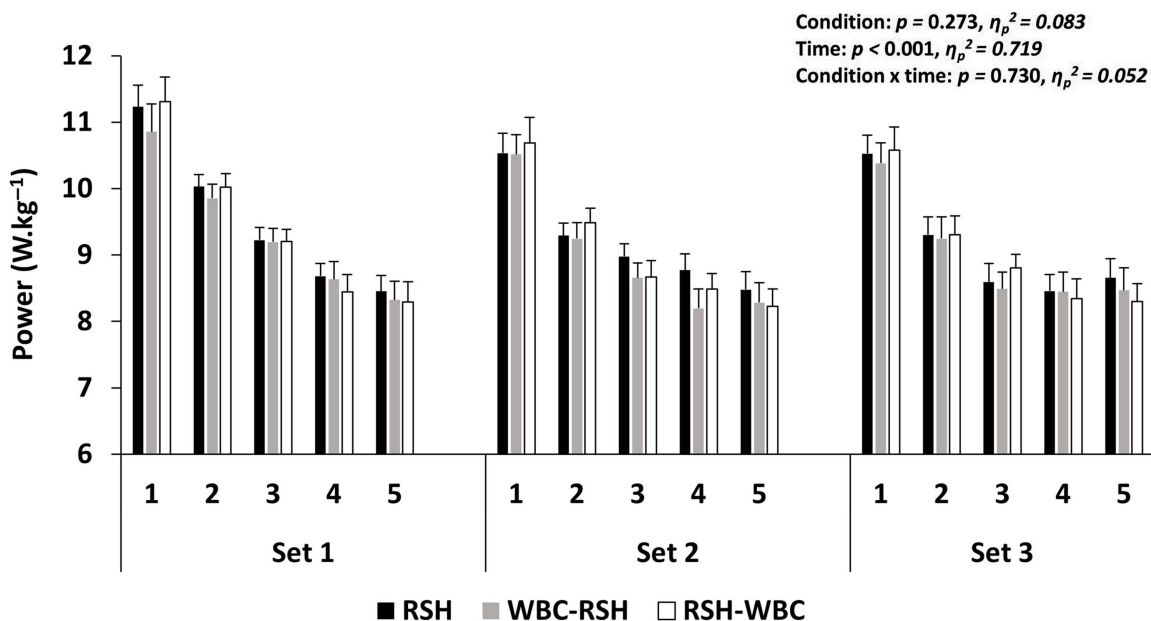


Figure 2 — Changes in mean ($\pm$ standard error) power during each sprint of RSH sessions. The decline in power output from sprint 1 to sprint 5 during each set was similar under the 3 conditions. RSH indicates repeated-sprint training in hypoxia; RSH-WBC, whole-body cryotherapy post-RSH; WBC-RSH, whole-body cryotherapy pre-RSH; η_p^2 , partial eta squared.

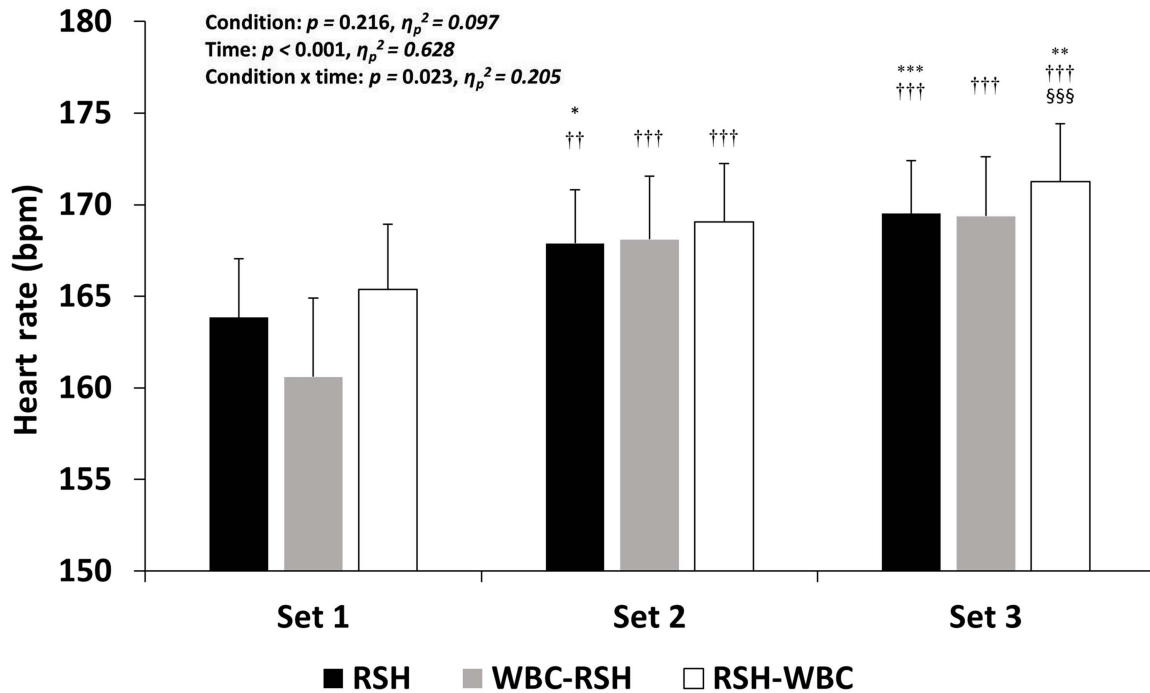


Figure 3 — Evolution of the mean ($\pm$ standard error) heart rate during the sets of RSH sessions. RSH indicates repeated-sprint training in hypoxia; RSH-WBC, whole-body cryotherapy post-RSH; WBC-RSH, whole-body cryotherapy pre-RSH; η_p^2 , partial eta squared. * $P < .05$, ** $P < .01$, *** $P < .001$ for differences with RSH set 1. †† $P < .01$ and ††† $P < .001$ for differences with WBC-RSH set 1; §§§ $P < .001$ for differences with RSH-WBC set 1.

[0.4] arbitrary units). The recovery perception measured at D and D + 1 is presented in Figure 5. Both conditions WBC-RSH and RSH-WBC were associated with a recovery perception improvement compared to RSH ($P = .011$, $d = 0.428$ [0.008 to 0.848]; $P < .001$, $d = 0.642$ [0.165 to 1.119], respectively).

Sleep Quality

HRV sleep indices and parameters for sleep quality assessment using actimetry and subjective approaches are presented in Table 2. Interestingly, nocturnal ActiGraphy results showed a significantly lower vector magnitude on the night after RSH-WBC compared to CONT ($P = .018$, $d = -0.565$ [-1.361 to 0.231]) and WBC-RSH ($P = .021$, $d = -0.431$ [-1.205 to 0.344]); without any effect on the other parameters (WASO, total time spent in bed, total sleep time, and sleep efficiency). A significant increase was observed in RSH compared to CONT and RSH-WBC for RMSSD ($P = .021$, $d = 0.925$ [-0.098 to 1.949]; $P = .006$, $d = 1.092$ [0.010 to 2.174], respectively) and HF ($P = .010$, $d = 1.184$ [-0.023 to 2.391]; $P = .002$, $d = 1.419$ [0.125 to 2.712], respectively).

Discussion

This study aimed to assess the acute psychophysiological effects of combining WBC and RSH during exercise, postexercise recovery and sleep quality. The main findings were that (1) the Δ [O₂HB] and Δ TSI amplitudes were attenuated at the end of the exercise in WBC-RSH without modifying the power output; (2) WBC improved recovery perception; and (3) sleep quality was positively influenced by RSH-WBC, with a decrease in the amount of movement.

Physiological Responses During Exercise

The results indicated that power output and S_{dec} during exercise did not differ between the 3 conditions (Figure 2). As already well-described,^{11,27} there was a significant decrease in power output between the first and last sprints of each set, along with an increase in [La⁻] and a decrease in TSI values. This reduction in TSI, more important in RSH than RSN,²⁷ was linked to an increase in [HHb] during sprints and a reduction in muscle reoxygenation during recovery periods, which was correlated with a reduction in mechanical power production.^{9,28} Interestingly, in the present study despite similar power output, Δ TSI calculated from baseline value, and used as an indicator of muscle oxygenation,¹⁰ was attenuated for WBC-RSH in set 3 compared to RSH-WBC (Figure 4). Δ TSI attenuation coincided with less deoxygenation during exercise, and was in accordance with Δ [O₂HB] reduced, observed in same time (Figure 4). This result was consistent with our first hypothesis, that WBC-RSH improves muscle oxygenation during hypoxic exercise. This hypothesis was based on the fact that cold-induced vasodilatation was observed (5–10 min) after cold exposure,²⁹ allowing peripheral blood circulation. The TSI maintenance had already been demonstrated in previous studies involving WBC exposure prior to exercise in normoxia.¹⁵ The enhanced oxygenation of the working muscles then led to a performance improvement during subsequent exercise, for example, time-trial effort in normoxia.^{15,30} Performance improvement in such an effort was linked to the capacity to maintain intensity over a prolonged period. In the present study, no change in performance was reported. Although speculative, it can be hypothesized that if participants had been asked to perform a maximum number of sprints to exhaustion (with a fixed minimal pedaling frequency for example, as described in Faiss et al³), performance would have been greater for WBC-RSH.

Table 1 Variation in HRV Indices and Blood Lactate During Test Sessions

Indices/conditions	Period				ANOVA P: C T C × T	η_p^2 : C T C × T
	HRV ₁	HRV ₂	HRV ₃	HRV ₄		
HR (% of HRV ₁)						
RSH	100	96.6 (5.1)	153 (23.4)	131 (23.1)	.300	.075
WBC-RSH	100	89.1 (4.5)	151 (19.4)	131 (17.1)	<.001	.883
RSH-WBC	100	94.4 (5.9)	152 (18.9)	123 (16.5)	.055	.174
RR (% of HRV ₁)						
RSH	100	103 (5.4)	66.4 (9.2)	78.1 (12.9)	.182	.113
WBC-RSH	100	112 (5.7)***††	67.3 (8.7)	77.6 (9.8)	<.001	.928
RSH-WBC	100	106 (6.4)	66.5 (8.4)	82.3 (11.0)	.002	.308
RMSSD (% of HRV ₁)						
RSH	100	114 (35.8)	25.9 (34.2)	49.8 (45.4)	.359	.066
WBC-RSH	100	125 (35.1)	14.4 (12.8)	46.7 (44.5)	<.001	.852
RSH-WBC	100	111 (30.0)	14.0 (14.0)	40.6 (27.1)	.316	.075
SDNN (% of HRV ₁)						
RSH	100	109 (25.1)	33.7 (15.2)	59.4 (40.5)	.135	.125
WBC-RSH	100	105 (25.1)	22.7 (15.2)	51.7 (40.5)	<.001	.853
RSH-WBC	100	99.0 (23.2)	20.6 (15.1)	45.8 (27.2)	.552	.045
SD1 (% of HRV ₁)						
RSH	100	114 (35.8)	14.4 (12.8)	44.9 (43.1)	.401	.059
WBC-RSH	100	120 (39.5)	25.9 (34.2)	49.8 (45.4)	<.001	.854
RSH-WBC	100	116 (26.0)	14.0 (13.9)	41.1 (27.8)	.747	.029
SD2 (% of HRV ₁)						
RSH	100	107 (35.7)	36.0 (25.9)	53.0 (38.6)	.360	.062
WBC-RSH	100	92.1 (26.9)	25.3 (16.1)	62.8 (45.0)	<.001	.835
RSH-WBC	100	100 (24.6)	24.0 (18.3)	49.4 (29.2)	.170	.108
LF (% of HRV ₁)						
RSH	100	186 (239)	26.5 (31.7)	90.1 (134)	.115	.134
WBC-RSH	100	80.0 (84.3)	13.0 (18.3)	60.2 (100)	<.001	.481
RSH-WBC	100	102 (73.9)	8.7 (14.1)	32.4 (37.0)	.129	.133
HF (% of HRV ₁)						
RSH	100	138 (105)	9.6 (21.6)	47.1 (68.1)	.348	.068
WBC-RSH	100	192 (177)	3.9 (5.8)	33.6 (46.5)	<.001	.679
RSH-WBC	100	124 (48.1)	3.8 (5.8)	35.2 (57.0)	.022	.096
	[La ⁻] ₁	[La ⁻] ₂	[La ⁻] ₃	[La ⁻] ₄		
Blood lactate, mmol·L ⁻¹						
RSH	5.4 (2.0)	15.0 (3.9)	9.4 (4.2)	5.0 (2.5)	.227	.094
WBC-RSH	4.7 (1.4)	15.2 (4.3)	9.8 (4.5)	4.9 (2.4)	<.001	.877
RSH-WBC	5.4 (1.5)	15.7 (3.4)	10.8 (4.6)	5.3 (2.9)	.603	.046

Abbreviations: η_p^2 , partial eta squared; ANOVA, analysis of variance; C, condition; C × T, condition × time interaction; HF, high-frequency spectral bands; HR, heart rate; HRV, heart-rate variability; LF, low-frequency spectral bands; RMSSD, root-mean-square successive difference; RR, time interval between beat-to-beat; RSH, repeated-sprint training in hypoxia; RSH-WBC, whole-body cryotherapy post-RSH; SD1, SD of instantaneous beat-to-beat RR interval variability; SD2, SD of long-term beat-to-beat RR interval variability; SDNN, SD of normal to normal R-R intervals; T, time; WBC-RSH, whole-body cryotherapy pre-RSH.

Values in bold for significant main effects. *** $P < .001$ for differences with RSH. †† $P < .01$ for differences with RSH-WBC.

WBC has been shown to facilitate increased [La⁻] clearance compared to passive recovery after training in normoxia, promoting a greater ability to repeat maximal aerobic and anaerobic performance.¹⁴ This was not supported by our data, since WBC exposure did not modify [La⁻] during RSH and [La⁻] removal after RSH. It is possible that the active recovery of 10 minutes at the end

of the experimental protocol (Figure 1), mainly influenced the [La⁻] clearance compared to the WBC.

WBC has been shown to influence HRV when completed before time-to-exhaustion effort; performance improvement was observed, and this was accompanied by a vagal drive enhancement at rest before and after exercise.³⁰ Our HRV results aligned with

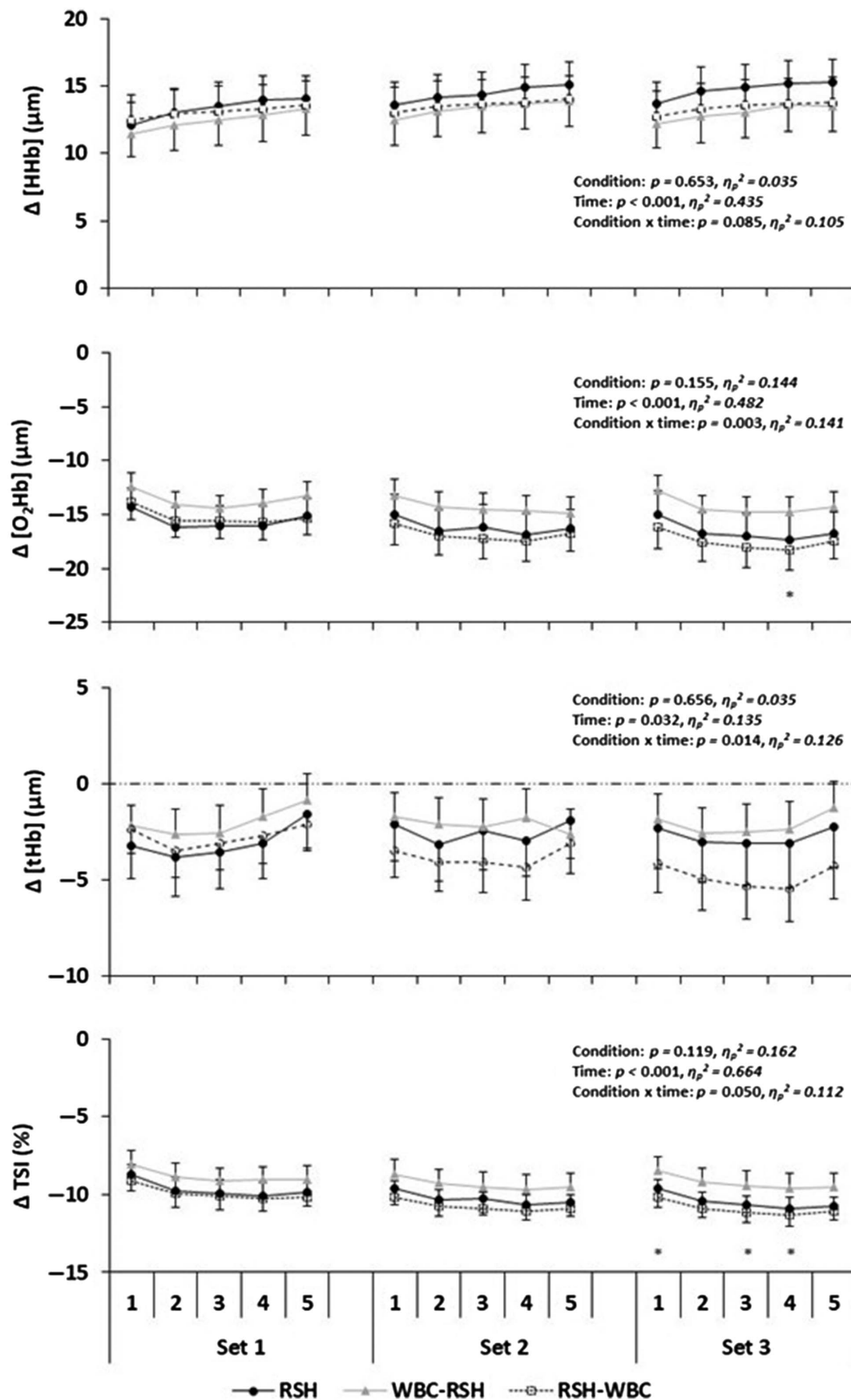


Figure 4 — Muscle concentration changes (Δ) in [HHb], [O₂Hb], [tHb], and TSI measured by near-infrared spectroscopy for each sprint of RSH sessions. Mean ($\pm$ standard error) changes are expressed in micrometers from the stabilization period performed before RSH set to 0 μm . [HHb] indicates deoxyhemoglobin; [O₂Hb], oxyhemoglobin; RSH, repeated-sprint training in hypoxia; RSH-WBC, whole-body cryotherapy post-RSH; [tHb], total hemoglobin; TSI, tissue saturation index; WBC-RSH, whole-body cryotherapy pre-RSH; η_p^2 , partial eta squared. * $P \leq .05$ for differences between WBC-RSH and RSH-WBC.

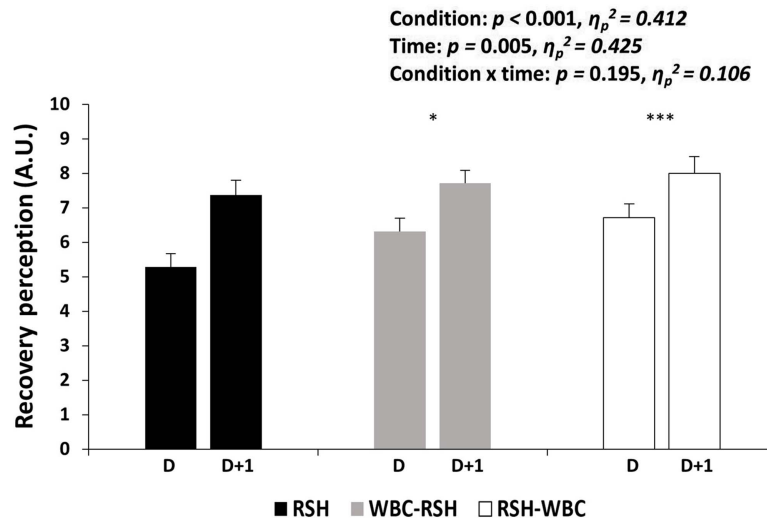


Figure 5 — Recovery perception measured at the end of the test session (D) and the next morning (D+1). A.U. indicates arbitrary units; RSH, repeated-sprint training in hypoxia; RSH-WBC, whole-body cryotherapy post-RSH; WBC-RSH, whole-body cryotherapy pre-RSH; η_p^2 , partial eta squared. * $P < .05$, *** $P < .001$ for differences with RSH.

Table 2 Night Heart-Rate-Variability Indices and Sleep Parameters

Indices	Condition				ANOVA <i>P</i>	η_p^2
	CONT	RSH	WBC-RSH	RSH-WBC		
HR (% of CONT)	100	97.0 (13.9)	99.3 (15.3)	106 (11.0)	.079	.196
RR (% of CONT)	100	105 (19.0)	102 (15.9)	95.2 (9.6)	.109	.192
RMSSD, %	100	134 (54.5)*††	117 (43.1)	93.8 (28.7)	.004	.329
SDNN (% of CONT)	100	113 (32.1)†	112 (26.2)†	93.0 (17.1)	.024	.245
SD1 (% of CONT)	100	134 (54.5)*††	117 (43.1)	93.8 (28.7)	.004	.329
SD2 (% of CONT)	100	99.2 (18.5)	110 (26.7)	93.0 (14.8)	.103	.169
LF (% of CONT)	100	110 (70.9)	130 (63.1)	89.6 (29.4)	.189	.133
HF (% of CONT)	100	172 (100)*††	126 (57.0)	85.6 (42.5)	.002	.364
WASO, min	57.5 (8.1)	53.1 (7.3)	53.9 (5.3)	49.4 (5.5)	.676	.029
VM, counts·min ⁻¹	85,195 (9198)	76,624 (9533)	80,671 (8115)	66,184 (6459)*§	.047	.098
TIB, min	515 (17.6)	513 (14.7)	503 (14.2)	493 (17.0)	.666	.034
TST, min	455 (16.1)	455 (13.3)	445 (11.5)	442 (16.7)	.824	.020
SE, %	88.4 (1.3)	88.8 (1.3)	88.6 (0.8)	89.7 (1.0)	.566	.039
Spiegel score, A.U.	22.6 (0.9)	22.4 (0.7)	21.8 (0.7)	21.9 (0.9)	.699	.031

Abbreviations: η_p^2 , partial eta squared; ANOVA, analysis of variance; A.U., arbitrary units; HF, high-frequency spectral bands; HR, heart rate; LF, low-frequency spectral bands; RMSSD, root-mean-square successive difference; RR, time interval between beat-to-beat; RSH, repeated-sprint training in hypoxia; RSH-WBC, whole-body cryotherapy post-RSH; SD1, SD of instantaneous beat-to-beat RR interval variability; SD2, SD of long-term beat-to-beat RR interval variability; SDNN, SD of normal to normal R-R intervals; SE, sleep efficiency; TIB, time in bed; TST, total sleep time; VM, vector magnitude; WASO, wake after sleep onset; WBC-RSH, whole-body cryotherapy pre-RSH; CONT, night recorded before the first test session and considered baseline.

Values in bold for significant main effects. * $P < .05$ for differences with CONT. † $P < .05$ and †† $P < .01$ for differences with RSH-WBC. § $P < .05$ for differences with WBC-RSH.

previous research: an increase in RR was observed between HRV₁ and HRV₂ for WBC-RSH compared to RSH ($P < .001$) and RSH-WBC ($P < .01$). Although HR was similar, on average, during exercise, the HR of WBC-RSH set 1 tended to be lower than RSH-WBC set 1 (Figure 3; $P = .09$, $d = -0.356$ [−0.859 to 0.146]). Thus, it appears that WBC-RSH allowed similar power output during RSH, with lower cardiac demand at the beginning of the effort. This may arise from parasympathetic activation before RSH due to WBC, as described before an all-out pedaling exercise in normoxia.³⁰

Recovery Perception

Despite similar physiological responses during exercise between conditions, both sessions with WBC presented higher perceived recovery than RSH condition (Figure 5; WBC-RSH: $P < .05$ and RSH-WBC: $P < .001$). Similar postexercise WBC benefits have been reported in motocross riders in normoxia.²⁰ Preexercise WBC is also known to have a positive effect on mood and perceived muscle soreness,³¹ partly due to its analgesic effect accompanied by the release of noradrenaline, which helps reduce pain and fatigue

perception and improve well-being.² Pain or fatigue perception is determined by both physiological and psychological influences and constitutes a relevant muscle recovery indicator. When performed before and after RSH, WBC improve recovery perception and could positively affect the performance/training quality in hypoxia by inculcating a positive state of mind in athletes.^{20,31}

Sleep Quality

Objective sleep quality was estimated as the amount of movement during sleep using a wrist ActiGraph and validated against laboratory polysomnography.¹⁶ The amount of movement characterized by vector magnitude was significantly reduced during the night following RSH-WBC compared to CONT ($P < .01$) and WBC-RSH ($P < .05$). High activity during nighttime is related to intermittent wakefulness during sleep, whereas prolonged episodes of immobility are associated with increasing sleep depth.³² Therefore, lower motor activity following an RSH-WBC night may be associated with deep sleep and a less agitated state. This result was observed in a previous study involving training in normoxia that coincided with an increase in subjective sleep quality.¹⁶ The significant reduction in the amount of movement after RSH-WBC may help participants to better recover via improved sleep quality.

This speculation should be substantiated by nocturnal HRV analysis after each test session. HRV indices measured during the first hour of sleep contradicted our second hypothesis since RSH presented a higher parasympathetic activity (RMSSD, HF) than CONT ($P < .05$) and RSH-WBC nights ($P < .001$). Past research has shown WBC's positive effect on parasympathetic activity during sleep after training in normoxia.¹⁶ Since parasympathetic activity is an important physiological biomarker of sleep stages and quality,³³ conditions involving WBC were expected to improve these parameters and potentially sleep quality after RSH.

Strengths and Limitations

To our knowledge, this is the first study that combines RSH and WBC. Using a randomized crossover design, we showed that WBC did not enhance power output but Δ TSI and Δ [O₂Hb] were attenuated in the last set of RSH. Therefore, this attenuation reflects a greater muscle oxygen availability despite the advancement of effort in hypoxia when this was preceded by WBC exposure. Moreover, in both conditions involving WBC the perceived recovery was improved.

Sleep quality analysis showed ambivalent results. Although validated, nocturnal actimetry is only a deduction from sleep quality. The use of polysomnography appears to be essential for the study of sleep architecture and post-RSH sleep quality and determine the real effect of combining WBC and RSH.

Practical Applications

The application of an acute WBC prior to RSH did not improve or decrease performance. However, a parasympathetic system activation prior and a small effect on HR at the beginning of the exercise in hypoxia were observed without power output degradation. In addition, as previously described in normoxia,¹⁵ WBC pre-RSH reduced the decrease in muscle oxygenation during the last set of exercise. Maintaining the oxygen supply to active muscles is beneficial during exercise and may accelerate subsequent recovery.

In a competitive hypoxic context, adding WBC could be a tactical approach for high-intensity activities, organized around several heats performed on the same day (sprint skiing and bicycle motocross race). Both effects may thus generate less fatigue and better performance in the decisive heats carried out at the end of the day. In training hypoxic context, WBC pre-RSH may be of limited interest, as it may reduce hypoxic stress and the subsequent adaptations. However, other studies are needed to conclude on the benefits of this combination for the performance improvement.

Like WBC post-RSH, the inclusion of WBC prior to RSH contributes to increased recovery perception after RSH. This improvement could thus contribute to the recovery and repetition of several high-intensity efforts with maximum psychological engagement. The study of the parameters related to sleep quality did not indicate any degradation attributed to RSH or any noticeable improvement related to WBC after RSH. It would be interesting to develop an experimental protocol that constitutes a complete training cycle to verify whether the fatigue accumulated during RSH sessions can be reduced via WBC, thereby improving sleep quality and posttraining recovery. Chronic improvement in recovery and sleep via WBC could result in greater performance enhancement at the end of the training cycle.

Conclusion

Repeated-sprint training in hypoxia (RSH) is an innovative training method that requires maximal intensity in a hypoxic environment and, therefore, may induce fatigue and/or reduced recovery. Inclusion of whole-body cryotherapy (WBC) in an RSH session did not modify the training quality (ie, power output) but allowed to maintain oxygen supply to the working muscles. In addition, sleep quality did not deteriorate after a single RSH session. WBC, particularly when performed after RSH, positively influenced recovery perception and sleep. To date, post-RSH WBC requires further investigation, particularly to confirm that it may optimize recovery during a complete RSH training cycle.

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